

Quiz 7

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer.

Problem 1. (10 points) Let $A = \begin{bmatrix} 2 & 4 & 0 & 5 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

(a) Find the dimension and a basis for the row space of A , $C(A^T)$.

$$\dim(C(A^T)) = 2, \text{ one basis } \left\{ \begin{bmatrix} 2 \\ 4 \\ 0 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \end{bmatrix} \right\}$$

(b) Find the dimension and a basis for the column space of A , $C(A)$.

$$\dim(C(A)) = 2, \text{ one basis } \left\{ \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

(c) Find the dimension and a basis for the nullspace of A , $N(A)$.

$$\dim(N(A)) = 4 - 2 = 2$$

$$\text{Solve } AX = 0$$

$$\text{let } x_2 = 1, x_4 = 0 \Rightarrow S_1 = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{let } x_2 = 0, x_4 = 1 \Rightarrow S_2 = \begin{bmatrix} -5/2 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow \text{one basis } \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -5/2 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

(d) Find the dimension and a basis for the left nullspace of A , $N(A^T)$.

$$\dim(N(A^T)) = 3 - 2 = 1$$

$$\text{Solve } A^T y = 0 \Rightarrow \begin{bmatrix} 2 & 0 & 0 \\ 4 & 0 & 0 \\ 0 & 1 & 0 \\ 5 & 2 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = 0 \Rightarrow \begin{cases} 2y_1 = 0 \\ 4y_1 = 0 \\ y_2 = 0 \\ 5y_1 + 2y_2 = 0 \end{cases} \Rightarrow y_1 = y_2 = 0$$

$$\Rightarrow y = c \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \text{ thus one basis } \left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$